2010376103 CSE4261 Assignment-9

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1 Training an autoencoder as a 2D feature generator and displaying CIFAR10 dataset's features

Github code link

An Autoencoder has been trained here. CIFER10 dataset has been used. Batch size is 512. Model has been trained to learn a 2-dimensional feature representation of each image for 20 epochs in PyTorch. Each class has been defined by unique color. The feature has been plotted in a scattered points, where resulting plot shows overlapping cluster of different color points with minimal class-wise separation. The autoencoder need to be tuned more. Figure 1 shows the 2D feature of different class levels in CIFER10 dataset.

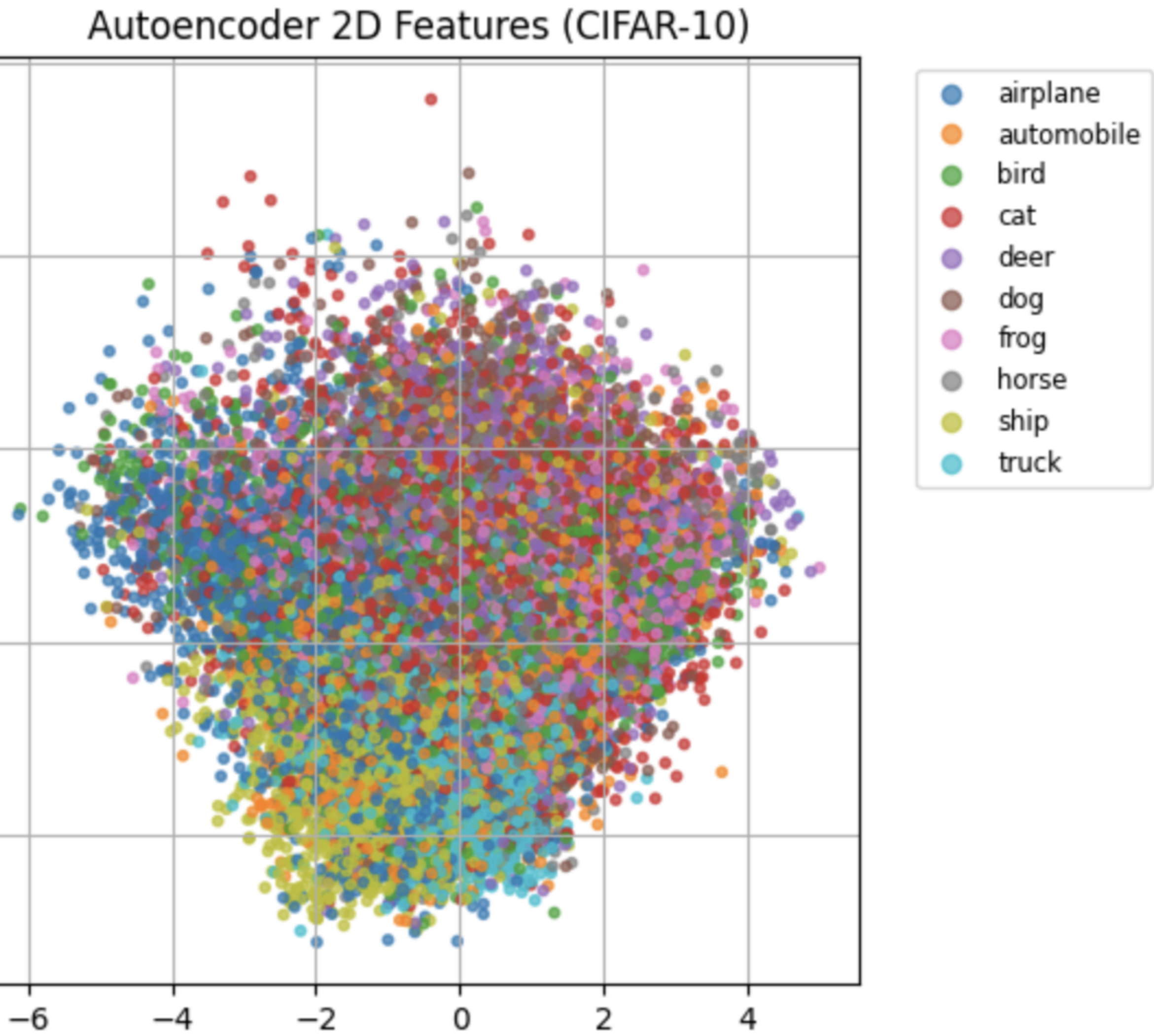
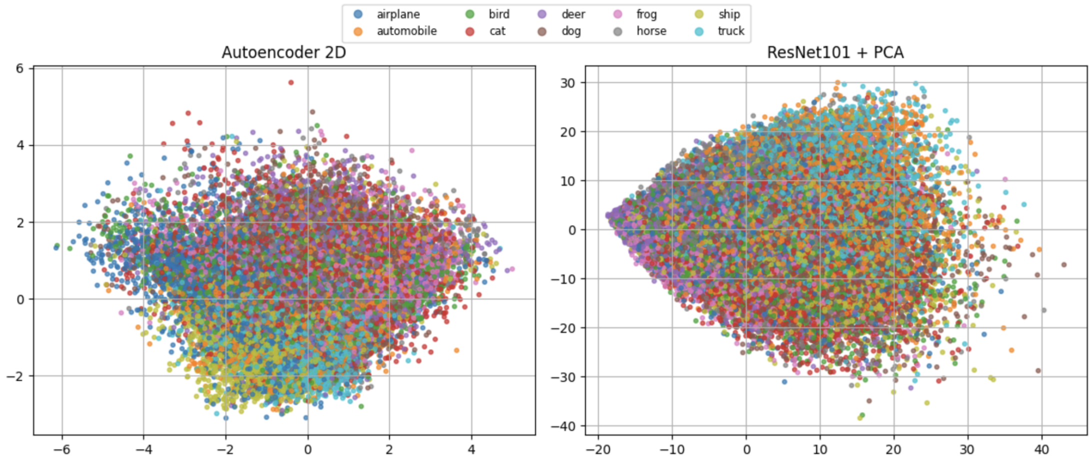


Figure 1: CIFAR10 dataset's features

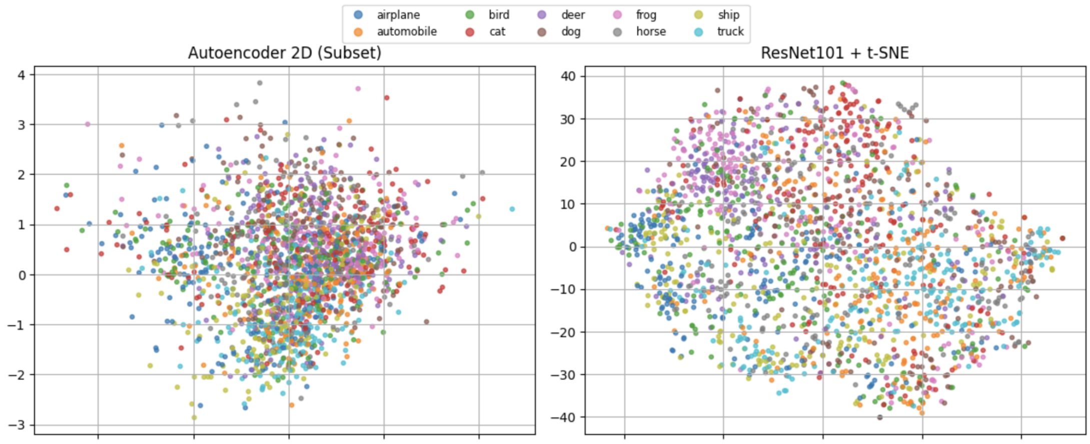
2 Comparing autoencoder generated features with features extracted by a pre-trained CNN and reduced by dimension reduction techniques like PCA, t-SNE.

Github code link

An Autoencoder has been trained here. CIFER10 dataset has been used. Batch size is 512. Model has been trained to learn a 2-dimensional feature representation of each image for 20 epochs in PyTorch. For Comparison pretrained model resnet101 has been used. PCA and t-SNE has has been used as dimension reduction technique. In the PCA comparison, both methods show dense feature distributions. t-SNE has shown better performance than PCA. Same class levels are close compared to AutoEncoder 2D feature. IN PCA same colors are close but overlapped. Figure 2 shows the Comparison of 2D features extracted by an autoencoder and a pretrained CNN using PCA and t-SNE for dimensionality reduction.



(a) Comparison using PCA



(b) Comparison using t-SNE

Figure 2: Comparison of 2D features extracted by an autoencoder and a pretrained CNN using PCA and t-SNE for dimensionality reduction.

3 training a denoising autoencoder for CIFAR10 dataset

Github code link

We used an autoencoder for denoising images. we have used CIFER10 dataset for training on batch size 128. We have implemented it in pytorch. The model has been run on 10 epochs and got upto Figure **??** shows the prediction result of two images in prediction map.

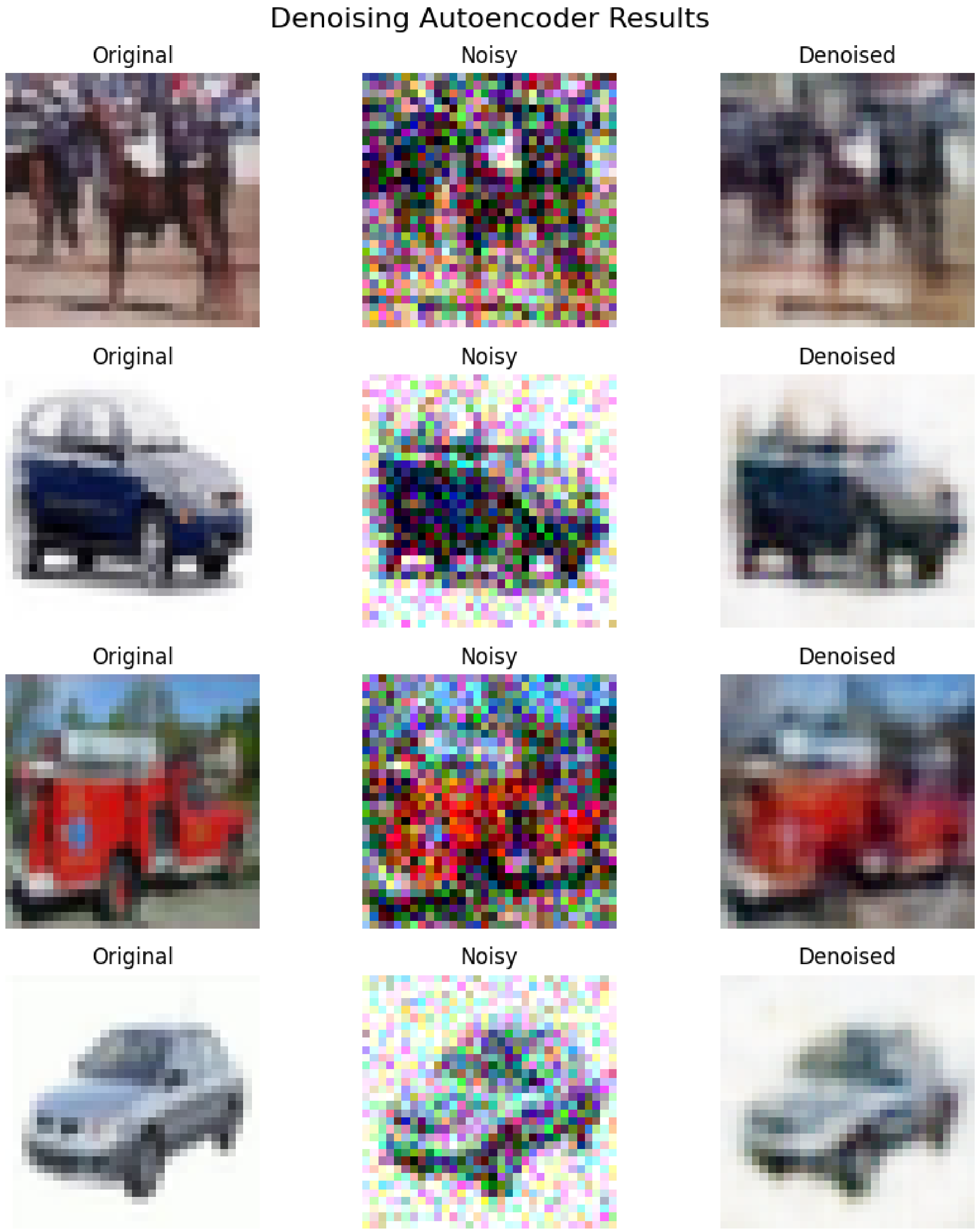


Figure 3: Original, noisy and denoised images

4 Training a CNN based CIFAR-10 classifier without any single-image data augmentation techniques

Github code link

A CNN based Classifier has been trained. CiFER10 dataset has been used on batch size 128. Only tensor transformation of images has been performed. No Augmentation has been performed to train the model. We got test Accuracy 74.26%. Figure 5 shows the images from datasets having different class.

Test Accuracy (no augmentation): 74.26%

Figure 4: Images Without Augmentation

5 Training a CNN based CIFAR-10 classifier with a single/multiple single-image data augmentation techniques

Github code link

A CNN based Classifier has been trained. CiFER10 dataset has been used on batch size 128. Augmentation has been performed to train the model. Random crop, random horizontal flip, color jitter, random rotation transformation has been performed in Augmentation. We got test Accuracy 75.18%. Figure 5 shows the images from datasets of a class with augmentation.

Test Accuracy : 75.18

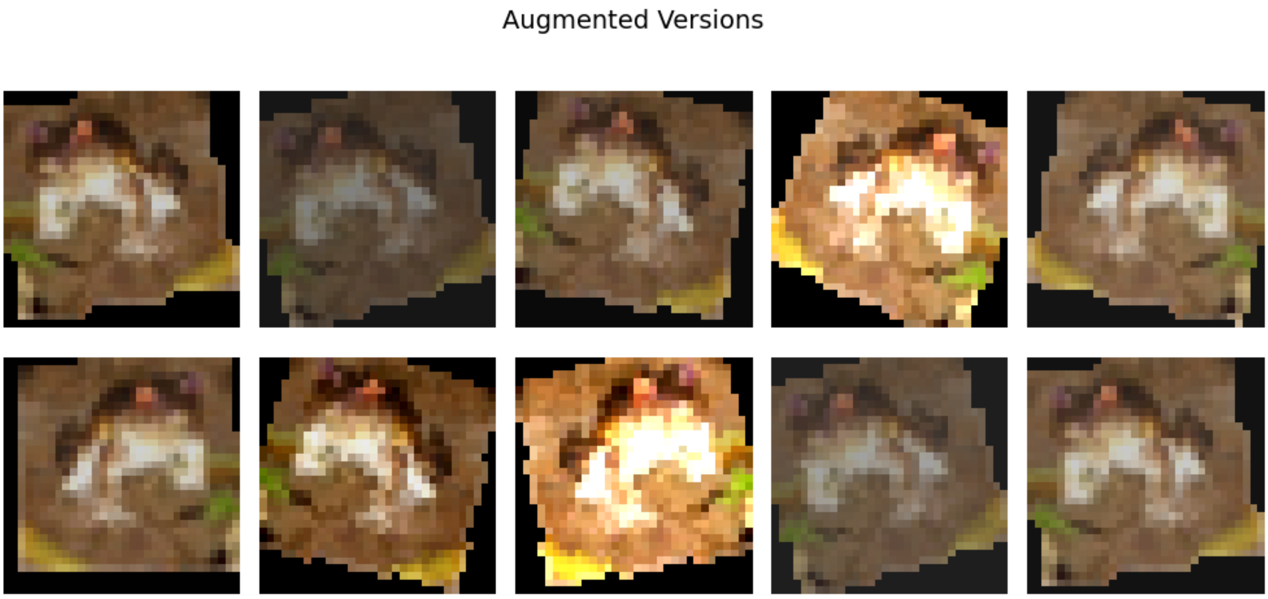


Figure 5: Images With Augmentation